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Slug: FACTCHECK SUPERIOR

Head: # Fact check: Is Lake Superior the fastest-warming lake on Earth?

Dek: *Superior's summer surface has warmed fast, faster than the air above it. But global surveys put it among the fastest-warming lakes, not alone at the top.*

Rating: **Needs context**

By Owen Brandt
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About 650 words, plus sources

The claim: "Lake Superior is warming faster than any lake on Earth." The line has been all over social media this summer, usually over a photo of the North Shore and rarely with a source.

Where it comes from: Most versions trace back to a 2007 study by two researchers at the University of Minnesota Duluth's Large Lakes Observatory. They found that Superior's surface water in summer had warmed about 2.5 degrees Celsius, or 4.5 degrees Fahrenheit, from 1979 to 2006. That was faster than the air over the region warmed in those years. A lake is expected to follow the air above it, not outrun it.

What it measured: The surface of the open lake in July, August and September, as recorded by federal weather buoys anchored offshore from spring to late fall. It did not measure the deep water, and it did not rank Superior against other lakes.

The explanation it offered is ice. Coming out of winter, Superior is 39 degrees or colder from top to bottom. Until the surface warms past 39 degrees, the temperature at which fresh water is heaviest, the lake keeps mixing and spreads the sun's heat through its whole depth. After that, a warm layer settles on top and heats quickly. Less ice means that layer forms sooner: about half a day earlier each year, the researchers estimated, or roughly two weeks over their record. A head start in June shows up as a warmer August.

What has been found since: Larger surveys came later, and they measured many lakes at once. A 2010 NASA study used satellite readings of summer nighttime surface temperatures for 167 large lakes and found most of them warming since 1985. In 2015, researchers from around the world pooled satellite and on-the-water records for 235 lakes. Summer surface temperatures rose an average of 0.34 degrees Celsius, about 0.6 degrees Fahrenheit, per decade from 1985 to 2009.

Superior sits near the fast end of those surveys. Its rate in the 2007 study works out to about 0.9 degrees Celsius per decade, nearly three times that average, though over different years. But it is not alone at the top. The fastest warmers turned up on several continents, and their order shifts with the years, the season and the method a study uses.

Why the years matter: Superior's summer surface swings hard with the winter before it. After the winter of 2014, when ice covered nearly all of the lake, the surface stayed cold well into summer. This past winter, ice peaked at 38 percent of the lake, against a long-term average of about 60. A 2014 study found that much of the rise in the 2007 record came in one jump after the warm El Niño winter of 1997-98, not as a steady climb. Change the years, and the trend changes.

What none of them measured: The deep lake. Superior averages about 480 feet deep and holds about 10 percent of the world's fresh surface water. Below the warm layer that forms each summer, the water stays near 39 degrees all year. A warmer summer surface is real, and it matters. It is not the same thing as a warmer lake from top to bottom.

Why it still matters: With or without a world ranking, less ice and a longer warm season at the surface reshape the lake. They change how much of the shore winter storms can reach, where lake trout and herring find cold water and how often algae blooms turn up along the South Shore.

Rating: Needs context. Superior's summer surface has warmed quickly, faster than the air above it, and the loss of winter ice is a big part of why. But "faster than any lake on Earth" goes beyond the research. Global surveys put Superior among the fastest-warming lakes, not alone at the top, and what they measured is the surface in summer, not the whole lake.

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SOURCES

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4. C.M. O'Reilly and others, Global Lake Temperature Collaboration: "Rapid and highly variable warming of lake surface waters around the globe," Geophysical Research Letters, 2015.
5. NOAA Great Lakes Environmental Research Laboratory: Superior ice cover, 1973 to 2026.
6. NOAA National Data Buoy Center: Lake Superior offshore buoy water temperatures.